P24

Knowledge, Attitude and Practice of Parents towards Childhood Vaccination in Perak during COVID-19 Endemic Phase

Chung Mei Ling; Lee Yi Yao; Teh Hooi Tian; Kirtana A/P Sivaraja; Pea Hui Ling; Ashvaani A/P Nadarajan
Hospital Slim River

Background & Objective

Context & Gap: Pre-COVID-19, Malaysian parents generally accepted vaccines, but post-pandemic data during the COVID-19 endemic phase remain limited. Persistent knowledge gaps, safety fears, and halal concerns still drive vaccine hesitancy

Objective: To determine parental Knowledge, Attitude, and Practice (KAP) towards childhood vaccination in Perak during the COVID-19 endemic phase.

Methodology

Design & sites: Cross-sectional study across 3 public hospitals in Perak (Hospital Raja Permaisuri Bainun, Hospital Teluk Intan, and Hospital Slim River).

Participants: Parents/caregivers aged ≥ 18 years with ≥ 1 child under 12 residing in Perak (N = 418). Minors and non-residents were excluded.

Data Collection: Self-administered via physical questionnaires or digital QR codes at outpatient pharmacies.

Instrument & Scoring: Pre-validated KAP questionnaire (10 Knowledge, 10 Attitude, 5 Practice items). Categorized via Bloom's cut-off (< 60% Poor, 60-79% Moderate, ≥ 80% Good).

Data Analysis: IBM SPSS v20. Descriptive statistics (medians/IQRs) and Chi-Square test for demographic associations (P < 0.05).

Participant Demographics & KAP Overview

Female (291, 69.6%)

Age 31-40 (227, 54.3%)

Malay (300, 71.8%)
Muslim (301, 72.0%)

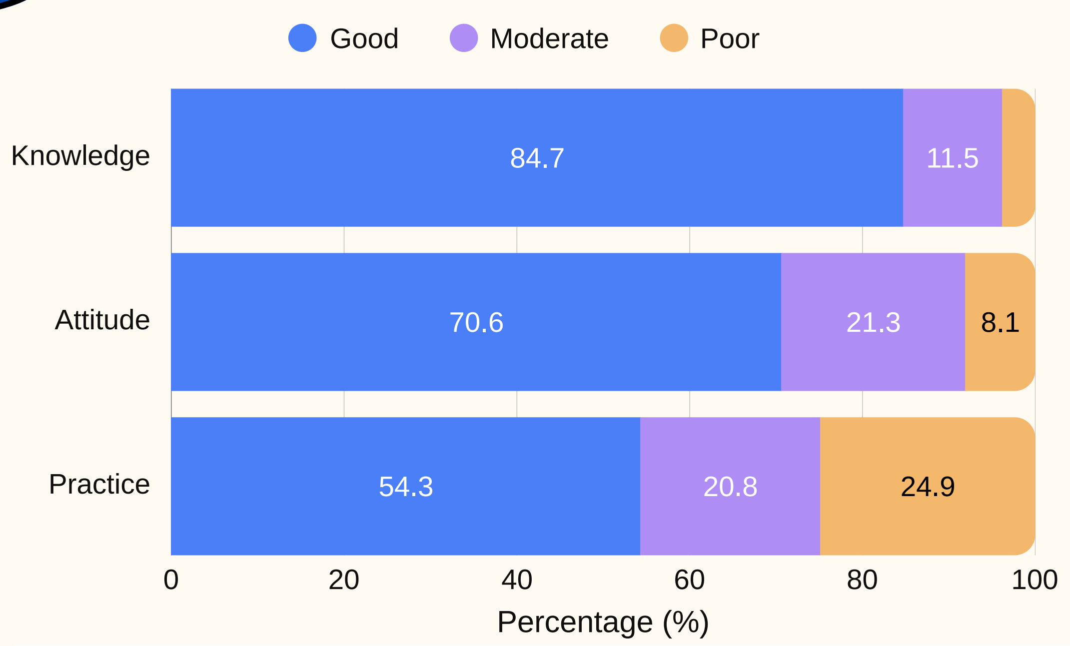
Private sector
(161, 38.5%)

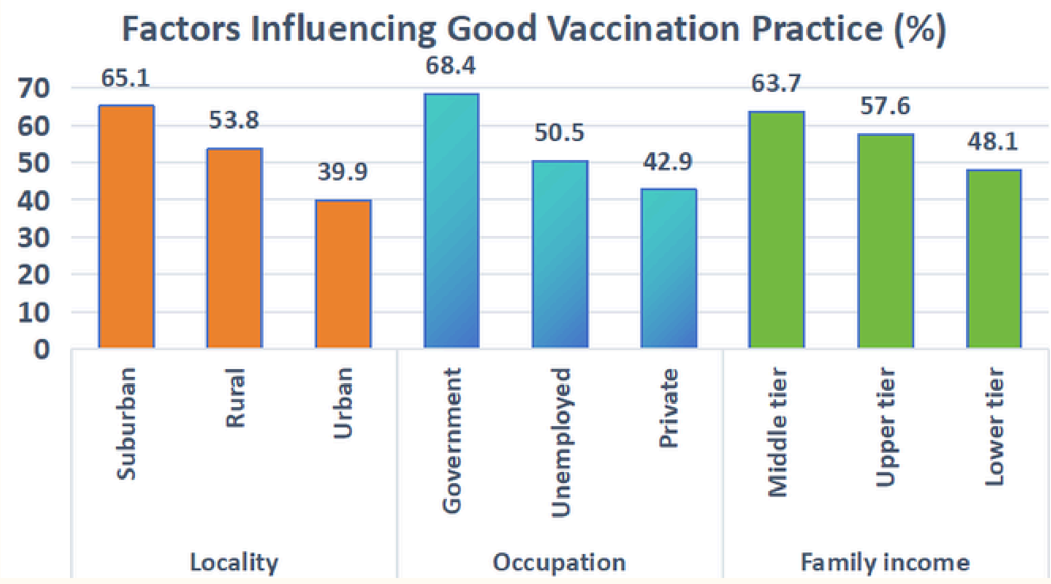
Suburban (195, 46.7%)

Family income
≤ RM4850 (239, 57.2%)

Characteristic	Key Breakdown (n, %)	Key Significant Factors (P < 0.05)
Knowledge	Good (354, 84.7%) Median : 9.0 (IQR 8-10)	Family income (P=0.023): Middle tier highest (91.1%) Occupation (P=0.045): Government highest (91.1%) Ethnicity (P=0.014): Chinese highest (88.0%)
Attitude	Good (295, 70.6%) Median: 9.0 (IQR 7-10)	Family income (P=0.008): Middle tier highest (76.0%)
Practice	Good (227, 54.3%) Median: 4.0 (IQR 3-5)	Locality (P<0.001): Suburban highest (65.1%) Occupation (P<0.001): Government highest (68,4%) Religion (P=0.001): Muslim highest (61.1%)

Overall Parental KAP Levels (Bloom's Cut-Off)





- **Practice Lag**: While knowledge was good (84.7%), actual vaccination adherence was poor, with only 54.3% achieving a good practice score.
- **Socioeconomic Drivers**: Higher income (RM4,851 - 10,970) and government employment provided significantly higher vaccine knowledge and practice levels (P < 0.05).
- **Urban Adherence Gap**: Practice adherence varied significantly by locality (65.1% suburban vs 39.9% urban, P < 0.001)

Discussion & Conclusion

- **The Knowledge-Practice Gap:** High knowledge (84.7%) and positive attitudes (70.6%) failed to translate into equivalent practice (54.3%).
- **Socioeconomic Determinants:** Higher income (RM 4,851 - 10,970) and government employment provided significantly greater vaccine literacy and schedule flexibility (P < 0.05).
- **Targeted Action Needed:** Public health strategies must shift from basic awareness to targeted interventions addressing halal concerns and logistical barriers in urban/low-income settings.

Limitation

Cross-sectional design limits causal inferences, and self-administered questionnaires may introduce self-reporting/selection bias.

Acknowledgement:
Sincere thanks to Dr. Sohail Aziz (MAHSA University), and the Hospital Directors, Heads of Pharmacy, and pharmacists at Hospital Slim River, Hospital Raja Permaisuri Bainun, and Hospital Teluk Intan.

Declaration
The authors declare no conflict of interest.

Reference:
1. Muhoza P, et al. MMWR. 2021;70:1495–1500.
2. Wong LP, et al. Vaccines. 2022;10:1790.
3. Aziz SO, et al. Int J Sci Eng Res. 2018;9:112–118.
4. Taber KS. Res Sci Educ. 2018;48:1273–1296.